

Hyunju Kim

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RESEARCH INTERESTS

Computational Neuroscience

Apply deep learning to understand neural dynamics, brain connectivity, and functional representations. Build brain-inspired deep learning models based on neural processing principles.

Graph Data Mining & Graph Algorithms

Design algorithms and AI models for analyzing large-scale, complex graph data.

EDUCATION

Georgia Institute of Technology

Ph.D. Student in Computational Science & Engineering. Advisor: Nabil Imam

Aug. 2024 –

Atlanta, GA, United States

KAIST (Korea Advanced Institute of Science and Technology)

M.S. in Artificial Intelligence. Advisor: Kijung Shin

Feb. 2022 – Feb. 2024

Daejeon, Republic of Korea

POSTECH (Pohang University of Science and Technology)

B.S. in Mathematics. Summa cum laude. GPA 3.96/4.3, Major GPA 4.02/4.3 (ranked 1st)

Feb. 2018 – Feb. 2022

Pohang, Republic of Korea

PUBLICATIONS

*: equal contributions C: Conference J: Journal P: Preprint

- [P3] Wiring motifs across functional systems of the fruit fly brain. *Under review*. **Hyunju Kim**, Nabil Imam
- [P2] Geometric Learning for Predicting Functional Connectivity on Folded Cortical Surfaces. *Under review*. **Hyunju Kim**, Michael Arcaro, Nabil Imam
- [P1] Geometric Constraints in the Development of Primate Extrastriate Visual Cortex. *Under review*. **Hyunju Kim**, Michael Arcaro, Nabil Imam (DOI: 10.64898/2026.02.04.703881)
- [J3] Four-Set Hypergraphlets for Characterization of Directed Hypergraphs. *TKDE*. Heechan Moon*, **Hyunju Kim***, Sunwoo Kim and Kijung Shin (DOI: 10.1109/TKDE.2026.3677137)
- [J2] Estimating Simplex Counts via Sampling. *The VLDB Journal*. **Hyunju Kim***, Heechan Moon*, Sunwoo Kim and Kijung Shin (DOI: 10.1007/s00778-024-00890-9)
- [J1] Hypergraph Motifs and their Extensions beyond Binary. *The VLDB Journal*. Geon Lee*, Seokbum Yoon*, Jihoon Ko, **Hyunju Kim** and Kijung Shin (DOI: 10.1007/s00778-023-00827-8)
- [C4] HyBrain: Spatiotemporal Hyperedges for EEG Seizure Detection and Prediction. *CIKM 2026*. **Hyunju Kim***, SheoYon Jhin*, Noseong Park, Nabil Imam
- [C3] DiffIM: Differentiable Influence Minimization with Surrogate Modeling and Continuous Relaxation. *AAAI 2025*. Junghun Lee, **Hyunju Kim**, Fanchen Bu, Jihoon Ko and Kijung Shin
- [C2] FlowerFormer: Empowering Neural Architecture Encoding using a Flow-aware Graph Transformer. *CVPR 2024*. Dongyeong Hwang, **Hyunju Kim**, Sunwoo Kim and Kijung Shin (DOI: 10.1109/CVPR52733.2024.00586)
- [C1] Characterization of Simplicial Complexes by Counting Simplexes beyond Four Nodes. *WWW 2023*. **Hyunju Kim**, Jihoon Ko, Fanchen Bu and Kijung Shin (DOI: 10.1145/3543507.3583332)

RESEARCH EXPERIENCES

EEG Analysis for Cognitive and Clinical Modeling

Georgia Tech

Sep. 2024 –

Drift-Diffusion Model (DDM) analysis of EEG signals for characterizing decision-making processes. EEG-based disease prediction leveraging neural biomarkers and deep learning methods [C4].

fMRI-driven Graph Modeling of Self-Organization Mechanisms in the Visual Cortex

Georgia Tech

Sep. 2024 –

Develop graph generation methods using both AI-driven approaches and rule-based algorithms. Design brain-inspired models by leveraging neural data visualization mechanisms [P1, P2].

Designed scalable algorithms for counting and characterizing higher-order structures (simplets, hypergraphlets, motifs) in simplicial complexes, hypergraphs, and directed hypergraphs.

Developed sampling-based estimation methods with provable accuracy guarantees [C1, J1, J2, J3].

Developed a null model maintaining performance rankings aligned with the original network while anonymizing users’ information.

Neural architecture performance prediction method using a flow-aware Graph Transformer.

Registered two South Korea Copyrights for Software (C-2022-055775, C-2023-047128).

Patent granted: KR 10-2980260; application KR 10-2024-0047575 [C2].

Supported by IITP grant funded by the Korea government (MSIT).

INDUSTRY EXPERIENCES

AWARDS & HONORS

ACM SIGIR Student Travel Grant	For attending CIKM 2026 [C4]	2026
SGA Conference Travel Grant by Georgia Tech	For attending CIKM 2026 [C4]	2026
KFAS (Korea Foundation for Advanced Studies) Scholarship	Field of Computer Science	2024 – 2029 (expected)
The PROUD POSTECHIAN MEMBER with the Highest Honor	Excellence Award (≈ \$400)	Feb. 2022
UGRP (Undergraduate Group Research Program)	Excellence Award (≈ \$4,000)	Jan. 2021
Presidential Science Scholarship by Republic of Korea	Full Tuition + Extra Fee (≈ \$8,000)	2020 – 2022
Jigok Scholarship by POSTECH	Full Tuition	2018 – 2020

PROFESSIONAL SERVICE & TEACHING

Reviewer	AAAI 2027, CIKM 2026, AAAI 2026, AAAI 2026 (AI Alignment Track), KDD 2026 (AI for Science Track)	
Head Teaching Assistant	Georgia Tech. CSE 6140: Computational Science & Engineering Algorithms	Fall 2026
URP Teaching Assistant	KAIST. Mentee: J.H. Lee → [C3]. Excellence Award.	Feb. 2023 – Oct. 2023
Lab Mentor	KAIST. Mentee: S.B. Yoon → [J1]	Aug. 2022 – Jun. 2023
Teaching Mentor & Tutor	POSTECH. Applied Linear Algebra, Analysis I, Programming and Problem Solving (7 semesters)	Fall 2019 – Fall 2021
Educational Volunteering Mentor	POSTECH. Online Mentoring & Korea Scholarship Foundation Summer Camp	Jul. 2019 – Dec. 2020

SKILLS

Programming Languages	C/C++, Python (DL: PyTorch), L ^A T _E X, Julia, R, JavaScript
	Korean (native), English (fluent: IELTS Overall 7.5; TOEFL – S 24, W 23)

MISCELLANEOUS

Identifying Fraudulent Products in Secondhand Trading Websites

Mar. 2021 – Jun. 2021

POSTECH. IMEN472 Statistical Data Mining term project (1st place, model case). Implemented web crawlers, machine learning algorithms and performed statistical analysis.

Personality Prediction AI and App Development

Apr. 2020 – Jan. 2021

POSTECH. UGRP Excellence Award (Advisor: Jaesik Park). Developed and distributed an app platform based on a DL-based model with high accuracy.

Korean I-Corps Program: Team Lutra

Sep. 2021 – Oct. 2021

POSTECH. Korean I-Corps program for supporting preliminary startups. Title: Development of a Personalized Restaurant and Cafe Recommendation Web Service.

Winner of POSTECH-KAIST Science War

Sep. 2020

POSTECH. Science Quiz Session. Mathematics representative.

Leadership Experiences

Feb. 2020 – Dec. 2021

POSTECH. The student council of the Department of Mathematics (Head of Academic Affairs & Public Relations). Orchestra club 'HanUIRim' (President & Conductor, Feb. 2020 – Dec. 2020).

Amateur Oboist & Classical Music Enthusiast

Guest player of several university orchestras (e.g., Sogang Univ., Sungkyunkwan Univ., Medical School Union, etc.). Principal oboe player of Yeodo Elementary School Orchestra European tour for promoting International Exposition Yeosu Korea 2012 (Venues: Musikverein, Mozarteum Grosser Saal, Slovak Radio Hall, World Trade Center Patria Hall, Rudolfinum Dvorak Hall, Lukaskirche). 9th Online Supporters, i.e., certified influencers, of Seoul Philharmonic Orchestra for promoting concerts.